How-To Library

Informatica

Address Verification Best Practices for the United Kingdom

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Abstract

This document tells you how to connect the elements in the United Kingdom address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in the United Kingdom.

Supported Versions

- Informatica Address Verification 6.3.0
- Informatica Address Verification (On-Premises) 6.3.0

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Introduction

Informatica Address Verification can validate and enhance addresses in the United Kingdom. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for the United Kingdom address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to analyze the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for validating United Kingdom addresses:

- ADDRESSCODELOOKUP

  Enter a UPRN or UDPRN code in address code lookup mode to receive the corresponding address.
- BATCH
- GEOCODETOADDRESS

- INTERACTIVE
- QUICKCAPTURE

  QuickCapture mode supports single-line data input. Use the SingleAddressLine element and the Country element to retrieve address suggestions.

Language and Character Set

The United Kingdom reference address databases are available in English.

The default character set for the United Kingdom addresses is Latin 1 - Western European.

UK Address Format and Address Examples

A typical United Kingdom address can contain some or all of the following types of information in the following order:

```
RECIPIENT
[SUB-BUILDING NAME OR NUMBER]
[BUILDING NAME]
[DEPENDENT THOROUGHFARE]
[HOUSE NUMBER AND THOROUGHFARE]
[DEPENDENT LOCALITY]
[LOCALITY (POST TOWN)]
[COUNTY]
[POSTCODE]
[COUNTRY]
```

Some elements are optional. A valid address must have a building name or house number or both.

The following table lists the address elements and examples for those elements.

Address Elements	Address Data
CONTACT	Mr. Walter Brown
HOUSE NUMBER AND STREET	49 Featherstone Street
LOCALITY (POST TOWN)	LONDON
POST CODE	EC1Y 8SY
COUNTRY	UNITED KINGDOM

Note: The address situates the post code on its own line below the locality name. The locality, post code, and country name are capitalized.

Key Address Elements

The following elements in United Kingdom addresses are keys for address verification:

Housenumber or building

Street

Locality

Post code

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map United Kingdom address information to the JSON address elements in Address Verification (On-Premises).

You enter the following input address:


```
{
        "Parameters": {
            "Mode": "Batch",
            "CountryDetermination": {},
            "DataSetDetermination": {},
            "CountrySets": [{
                    "Process": {
                        "AlternativeHandling": {},
                        "VerificationLevel": {
                            "PreferredVerificationLevel": "All",
                            "MinimumVerificationLevel": "All"
                        },
                        "Geocode": {
                            "SearchRadius": "50"
                        },
                        "Enrichments": {
                            "Geocoding": {
                                "Rooftop": false,
                                "ArrivalPoint": false,
                                "StreetCenter": false,
                                "PostalCodeCenter": false,
                                "LocalityCenter": false
                            },
                            "Global": false,
                            "CountrySpecific": false,
                            "CAMEO": false
                        }
                    },
                    "Input": {
                        "SingleAddressLineDataType": "UndelimitedSingleLine",
                        "SingleAddressLineDelimiter": "Comma"
                    },
                    "Result": {
                        "DeliverAdditionalSuggestions": "ForPostalRelevantElements",
                        "NumericRangeExpansion": {
                            "RangesToExpand": "None",
                            "RangeExpansionType": "Flexible"
                        }
                    },
                    "OutputDetail": {
                        "PreformattedData": {
                            "PostalFormattedAddressLines": true,
                            "PostalDeliveryAddressLines": true,
                            "PostalRecipientLines": true,
                            "PostalLocalityLine": true,
                            "StreetWithNumber": false,
                            "NumberAndSubBuilding": false,
                            "SingleAddressLine": true,
                            "SingleAddressLineDelimiter": "Semicolon"
                        },
                        "SubItems": true,
                        "ElementStatus": true,
                        "OutputStatusForEmptyElements": true
                    },
                    "Standardizations": [{
                            "Default": {
                                "PreferredScript": {
                                    "Script": "NativePrimary",
                                    "TransliterationType": "Default",
                                    "LimitLatinCharacters": "NoLimit"
                                },
                                "StandardizeInvalidAddresses": false,
                                "FormatWithCountry": false,
```

```
                                "CountryNameLanguage": "ENG",
                                "CountryCodeType": "ISO2",
                                "MaxItemLength": 255,
                                "Casing": "PostalAdmin",
                                "DescriptorLength": "Database",
                                "AliasHandling": "PostalAdmin"
                            }
                        }
                    ]
                }
            ]
        },
        "IO": {
            "Inputs": [{
                    "AddressElements": {
                        "Building": [{
                                "Value": "THE MOORINGS"
                            }
                        ],
                        "Street": [{
                                "Value": "MILL HOO"
                            }
                        ],
                        "Locality": [{
                                "Value": "WOODBRIDGE"
                            }, {
                                "Value": "ALDERTON"
                            }
                        ],
                        "PostalCode": [{
                                "Value": "IP123DA"
                            }
                        ],
                        "AdministrativeDivision": [{
                                "Value": "SUFFOLK"
                            }, {
                                "Value": "SUFFOLK"
                            }, {
                                "Value": "SUFFOLK"
                            }
                        ],
                        "Country": "GBR"
                    }
                }
            ]
        }
    }
```

Informatica Address Verification (On-Premises) returns the following result:

```
{
  "Parameters": {
    "Mode": "Batch",
    "CountryDetermination": {},
    "DataSetDetermination": {},
    "CountrySets": [
      {
        "Process": {
          "AlternativeHandling": {},
          "VerificationLevel": {
            "PreferredVerificationLevel": "All",
            "MinimumVerificationLevel": "All"
          },
          "Geocode": { "SearchRadius": "50" },
          "Enrichments": {
            "Geocoding": {
              "Rooftop": false,
              "ArrivalPoint": false,
              "StreetCenter": false,
              "PostalCodeCenter": false,
              "LocalityCenter": false
```

```
            },
            "Global": false,
            "CountrySpecific": false,
            "CAMEO": false
          }
        },
        "Input": {
          "SingleAddressLineDataType": "UndelimitedSingleLine",
          "SingleAddressLineDelimiter": "Comma"
        },
        "Result": {
          "DeliverAdditionalSuggestions": "ForPostalRelevantElements",
          "NumericRangeExpansion": {
            "RangesToExpand": "None",
            "RangeExpansionType": "Flexible"
          }
        },
        "OutputDetail": {
          "PreformattedData": {
            "PostalFormattedAddressLines": true,
            "PostalDeliveryAddressLines": true,
            "PostalRecipientLines": true,
            "PostalLocalityLine": true,
            "StreetWithNumber": false,
            "NumberAndSubBuilding": false,
            "SingleAddressLine": true,
            "SingleAddressLineDelimiter": "Semicolon"
          },
          "SubItems": true,
          "ElementStatus": true,
          "OutputStatusForEmptyElements": true
        },
        "Standardizations": [
          {
            "Default": {
              "PreferredScript": {
                "Script": "NativePrimary",
                "TransliterationType": "Default",
                "LimitLatinCharacters": "NoLimit"
              },
              "StandardizeInvalidAddresses": false,
              "FormatWithCountry": false,
              "CountryNameLanguage": "ENG",
              "CountryCodeType": "ISO2",
              "MaxItemLength": 255,
              "Casing": "PostalAdmin",
              "DescriptorLength": "Database",
              "AliasHandling": "PostalAdmin"
            }
          }
        ]
      }
    ]
  },
  "IO": {
    "Inputs": [
      {
        "AddressElements": {
          "Building": [ { "Value": "THE MOORINGS" } ],
          "Street": [ { "Value": "MILL HOO" } ],
          "Locality": [ { "Value": "WOODBRIDGE" }, { "Value": "ALDERTON" } ],
          "PostalCode": [ { "Value": "IP123DA" } ],
          "AdministrativeDivision": [
            { "Value": "SUFFOLK" }, { "Value": "SUFFOLK" }, { "Value": "SUFFOLK" }
          ],
          "Country": "GBR"
        }
      }
    ],
    "Outputs": [
      {
```

```
"ResultInfo": {
  "ProcessStatus": "V",
  "ProcessModeUsed": "Batch",
  "DeliverAdditionalSuggestions": "No",
  "ResultCount": 1,
  "ResultCountOverflow": false,
  "ResultCountries": [ "GBR" ]
},
"Results": [
  {
    "Variants": [
      {
        "StatusValues": {
          "AddressType": "B",
          "LanguageISO3": "ENG",
          "UsedVerificationLevel": "All",
          "MatchPercentage": "100.00",
          "Script": "Native Primary",
          "AddressCount": "",
          "ResultQuality": 6
        },
        "AddressElements": {
          "Building": [
            {
              "Value": "THE MOORINGS",
              "SubItems": { "Name": "THE MOORINGS" },
              "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
              }
            }
          ],
          "Street": [
            {
              "Value": "MILL HOO",
              "SubItems": { "Name": "MILL HOO" },
              "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
              }
            }
          ],
          "Locality": [
            {
              "Value": "WOODBRIDGE",
              "SubItems": { "Name": "WOODBRIDGE" },
              "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
              }
            },
            {
              "Value": "ALDERTON",
              "SubItems": { "Name": "ALDERTON" },
              "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
              }
            }
          ],
          "PostalCode": [
            {
              "Value": "IP12 3DA",
              "SubItems": { "Base": "IP123DA" },
              "Status": {
                "MatchStatus": "7",
                "ResultStatus": "2",
```

```
                    "PostalRelevance": "2"
                  }
                }
              ],
              "AdministrativeDivision": [
                {
                  "Value": "SUFFOLK",
                  "Variants": { "Extended": "SUFFOLK" },
                  "Status": {
                    "MatchStatus": "8",
                    "ResultStatus": "1",
                    "PostalRelevance": "0"
                  }
                },
                {
                  "Value": "SUFFOLK",
                  "Variants": { "Extended": "SUFFOLK", "ISO": "SFK" },
                  "Status": {
                    "MatchStatus": "8",
                    "ResultStatus": "1",
                    "PostalRelevance": "0"
                  }
                },
                {
                  "Value": "SUFFOLK",
                  "Variants": { "Extended": "SUFFOLK" },
                  "Status": {
                    "MatchStatus": "8",
                    "ResultStatus": "1",
                    "PostalRelevance": "0"
                  }
                }
              ],
              "Country": [ { "Code": "GB", "Name": "UNITED KINGDOM" } ]
            },
            "PreformattedData": {
              "SingleAddressLine": {
                "Value": "THE MOORINGS;MILL HOO;ALDERTON;WOODBRIDGE;IP12 3DA"
              },
              "PostalDeliveryAddressLines": [
                { "Value": "THE MOORINGS" }, { "Value": "MILL HOO" }
              ],
              "PostalFormattedAddressLines": [
                { "Value": "THE MOORINGS" }, { "Value": "MILL HOO" }, { "Value":
"ALDERTON" },
                { "Value": "WOODBRIDGE" }, { "Value": "IP12 3DA" }
              ],
              "PostalLocalityLine": { "Value": "WOODBRIDGE" }
            }
          }
        ]
      }
    ]
  }
}
```


Input and Result XML in Address Verification

The following example shows how you can map United Kingdom address information to the JSON address elements in Address Verification.

You submit the following request:


```
{
    "Login": "*****",
    "Password": "*****",
    "UseTransactions": "Test",
```

```
"CampaignId": "",
"Request": {
    "Parameters": {
        "Mode": "Batch",
        "CountryDetermination": {},
        "DataSetDetermination": {},
        "CountrySets": [{
                "Process": {
                    "AlternativeHandling": {},
                    "VerificationLevel": {
                        "PreferredVerificationLevel": "All",
                        "MinimumVerificationLevel": "All"
                    },
                    "Geocode": {
                        "SearchRadius": "50"
                    },
                    "Enrichments": {
                        "Geocoding": {
                            "Rooftop": false,
                            "ArrivalPoint": false,
                            "StreetCenter": false,
                            "PostalCodeCenter": false,
                            "LocalityCenter": false
                        },
                        "Global": false,
                        "CountrySpecific": false,
                        "CAMEO": false
                    }
                },
                "Input": {
                    "SingleAddressLineDataType": "UndelimitedSingleLine",
                    "SingleAddressLineDelimiter": "Comma"
                },
                "Result": {
                    "DeliverAdditionalSuggestions": "ForPostalRelevantElements",
                    "NumericRangeExpansion": {
                        "RangesToExpand": "None",
                        "RangeExpansionType": "Flexible"
                    }
                },
                "OutputDetail": {
                    "PreformattedData": {
                        "PostalFormattedAddressLines": true,
                        "PostalDeliveryAddressLines": true,
                        "PostalRecipientLines": true,
                        "PostalLocalityLine": true,
                        "StreetWithNumber": false,
                        "NumberAndSubBuilding": false,
                        "SingleAddressLine": true,
                        "SingleAddressLineDelimiter": "Semicolon"
                    },
                    "SubItems": true,
                    "ElementStatus": true,
                    "OutputStatusForEmptyElements": true
                },
                "Standardizations": [{
                        "Default": {
                            "PreferredScript": {
                                "Script": "NativePrimary",
                                "TransliterationType": "Default",
                                "LimitLatinCharacters": "NoLimit"
                            },
                            "StandardizeInvalidAddresses": false,
                            "FormatWithCountry": false,
                            "CountryNameLanguage": "ENG",
                            "CountryCodeType": "ISO2",
                            "MaxItemLength": 255,
                            "Casing": "PostalAdmin",
                            "DescriptorLength": "Database",
                            "AliasHandling": "PostalAdmin"
                        }
```

```
                        }
                    ]
                }
            ]
        },
        "IO": {
            "Inputs": [{
                    "AddressElements": {
                        "Building": [{
                                "Value": "THE MOORINGS"
                            }
                        ],
                        "Street": [{
                                "Value": "MILL HOO"
                            }
                        ],
                        "Locality": [{
                                "Value": "WOODBRIDGE"
                            }, {
                                "Value": "ALDERTON"
                            }
                        ],
                        "PostalCode": [{
                                "Value": "IP123DA"
                            }
                        ],
                        "AdministrativeDivision": [{
                                "Value": "SUFFOLK"
                            }, {
                                "Value": "SUFFOLK"
                            }, {
                                "Value": "SUFFOLK"
                            }
                        ],
                        "Country": "GBR"
                    }
                }
            ]
        }
    }
}
```

Informatica Address Verification returns the following response:

```
{
    "Status": "Ok",
    "StatusDescription": "No error",
    "TransactionInfo": {
        "TotalTransactionsCharged": 1,
        "TransactionPools": [
            {
                "Type": "BATCH",
                "TransactionsCharged": 1
            }
        ]
    },
    "Response": [
        {
            "ResultInfo": {
                "ProcessStatus": "V",
                "ProcessModeUsed": "Batch",
                "ResultCount": 1,
                "ResultCountOverflow": false,
                "ResultCountries": [
                    "GBR"
                ]
            },
            "Results": [
                {
                    "Variants": [
                        {
```

```
"StatusValues": {
    "AddressType": "B",
    "LanguageISO3": "ENG",
    "UsedVerificationLevel": "ALL",
    "MatchPercentage": "100.00",
    "Script": "Database",
    "AddressCount": "",
    "ResultQuality": 6
},
"AddressElements": {
    "Building": [
        {
            "Value": "THE MOORINGS",
            "SubItems": {
                "Name": "THE MOORINGS"
            },
            "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
            }
        }
    ],
    "Street": [
        {
            "Value": "MILL HOO",
            "SubItems": {
                "Name": "MILL HOO"
            },
            "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
            }
        }
    ],
    "Locality": [
        {
            "Value": "WOODBRIDGE",
            "SubItems": {
                "Name": "WOODBRIDGE"
            },
            "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
            }
        },
        {
            "Value": "ALDERTON",
            "SubItems": {
                "Name": "ALDERTON"
            },
            "Status": {
                "MatchStatus": "8",
                "ResultStatus": "1",
                "PostalRelevance": "2"
            }
        }
    ],
    "PostalCode": [
        {
            "Value": "IP12 3DA",
            "SubItems": {
                "Base": "IP123DA"
            },
            "Status": {
                "MatchStatus": "7",
                "ResultStatus": "2",
                "PostalRelevance": "2"
            }
```

```
                }
            ],
            "AdministrativeDivision": [
                {
                    "Value": "SUFFOLK",
                    "Variants": {
                        "Extended": "SUFFOLK"
                    },
                    "Status": {
                        "MatchStatus": "8",
                        "ResultStatus": "1",
                        "PostalRelevance": "0"
                    }
                },
                {
                    "Value": "SUFFOLK",
                    "Variants": {
                        "Extended": "SUFFOLK",
                        "ISO": "SFK"
                    },
                    "Status": {
                        "MatchStatus": "8",
                        "ResultStatus": "1",
                        "PostalRelevance": "0"
                    }
                },
                {
                    "Value": "SUFFOLK",
                    "Variants": {
                        "Extended": "SUFFOLK"
                    },
                    "Status": {
                        "MatchStatus": "8",
                        "ResultStatus": "1",
                        "PostalRelevance": "0"
                    }
                }
            ],
            "Country": [
                {
                    "Code": "GB",
                    "Name": "UNITED KINGDOM"
                }
            ]
        },
        "PreformattedData": {
            "SingleAddressLine": {
                "Value": "THE MOORINGS;MILL HOO;ALDERTON;WOODBRIDGE;IP12
3DA"
            },
            "PostalDeliveryAddressLines": [
                {
                    "Value": "THE MOORINGS"
                },
                {
                    "Value": "MILL HOO"
                }
            ],
            "PostalFormattedAddressLines": [
                {
                    "Value": "THE MOORINGS"
                },
                {
                    "Value": "MILL HOO"
                },
                {
                    "Value": "ALDERTON"
                },
                {
                    "Value": "WOODBRIDGE"
                },
```

```
                                    {
                                        "Value": "IP12 3DA"
                                    }
                                ],
                                "PostalLocalityLine": {
                                    "Value": "WOODBRIDGE"
                                }
                            }
                        }
                    ]
                }
            ]
        }
    ]
}
```


Enrichments for the United Kingdom Addresses

You can configure Informatica Address Verification to add enrichments to verified United Kingdom addresses.

The following table lists the types of enrichments that you can add to a United Kingdom address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid United Kingdom addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Country-Specific Address Enrichments	You can retrieve the following enrichments for valid United Kingdom addresses: - AddressKey. An eight-digit numeric code that maps to an address in the Postcode Address File (PAF) from the Royal Mail. - DeliveryPointSuffix. A two-character suffix that the Royal Mail assigns to a mailbox in a United Kingdom post code area. - DeliveryPointType. A single-character code that indicates whether the address points to a residence (R), a small organization (O), or a large organization (L). - OrganizationKey. A unique eight-digit numeric code that Royal Mail assigns to small organizations. - UDPRN. An eight-character code that uniquely identifies each postal address in the Royal Mail PAF database. - UPRN. A numeric code that uniquely identifies a land or property unit in the United Kingdom.
Geocoding	You can retrieve StreetCenter, ArrivalPoint, and Rooftop geocoordinates for valid United Kingdom addresses.
Reverse Geocoding	You can retrieve one or more addresses that match a set of geocoordinates in the United Kingdom when you submit the geocoordinates in GEOCODETOADDRESS mode.

Frequently Asked Questions

What is the maximum number of address lines in a United Kingdom address?

According to the Universal Postal Union specifications, the maximum number of address lines in a United Kingdom address is six. This does not include the name of the person to whom the mail is addressed or the country information line.

Is province information available for United Kingdom addresses?

Yes. In United Kingdom addresses, province information equates to county information. If a Postal County is available for the address, Informatica Address Verification returns the Postal County name in the AdministrativeDivision field.

Address Verification can return the county names regardless of whether an address requires county information. For example, the following address is valid without the county name:

```
Unit 1
Anderson Road
Woodford Green
IG8 8ET
United Kingdom
```

How does Address Verification handle county names in United Kingdom addresses?

The Royal Mail Postcode Address File (PAF) contains county names in postal, administrative, and traditional forms. Address Verification returns the county names in the AdministrativeDivision element.

The AdministrativeDivision element returns the values in an array. The first value in the array is the postal province, the second value is the administrative province, and the third value is the traditional province.

What is a dependent street in a United Kingdom address?

A dependent street in a United Kingdom address identifies a second-level street. Dependent street information depends on a main street information.

How important is house number information in a United Kingdom address?

A valid United Kingdom address contains a house number or a building name or both. House number information is not required if the address contains a building name. Similarly, the building name is not required if the address contains a house number.

Reference Address Databases

Informatica updates the reference address databases for United Kingdom every month. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the newsletter from Informatica.

The following table lists the reference databases that Informatica provides for United Kingdom addresses:

Database Name	Database Contents
GBR_ADV_ENR_CAM_000_6_1_0.MD6	CAMEO enrichments
GBR_ADV_ENR_GAP_000_6_1_0.MD6	Geocoding data, ArrivalPoint level
GBR_ADV_ENR_GRT_000_6_1_0.MD6	Geocoding data, Rooftop level
GBR_ADV_ENR_GST_000_6_1_0.MD6	Geocoding data, Standard level
GBR_ADV_ENR_EN1_000_6_1_0.MD6	Country-specific enrichments
GBR_ADV_RLK_ACL_000_6_1_0.MD6	Address code lookup data
GBR_ADV_RLK_GTA_000_6_3_0.MD6	Reverse Lookup - Geocode to address data

Database Name	Database Contents
GBR_ADV_VRF_BIA_000_6_3_0.MD6	Batch/Interactive verification data
GBR_ADV_VRF_QCP_000_6_3_0.MD6	QuickCapture verification data

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, check the following publications on the Informatica Documentation Portal:

- Address Verification (On-Premises) 6.3
- Address Verification 6.2

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